Series : WYXZ7

SET ~ 3

प्रश्न-पत्र कोड
Q.P. Code **56/7/3**

रोल नं.
Roll No.

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें।
Candidates must write the Q.P. Code on the title page of the answer-book.

(I) कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ **23** हैं।
Please check that this question paper contains **23** printed pages.

(II) प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें।
Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.

(III) कृपया जाँच कर लें कि इस प्रश्न-पत्र में **33** प्रश्न हैं।
Please check that this question paper contains **33** questions.

(IV) **कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में यथा स्थान पर प्रश्न का क्रमांक अवश्य लिखें।**
Please write down the Serial Number of the question in the answer-book at the given place before attempting it.

(V) इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है। प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा। 10.15 बजे से 10.30 बजे तक परीक्षार्थी केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे।
15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the candidates will read the question paper only and will not write any answer on the answer-book during this period.

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रसायन विज्ञान (सैद्धान्तिक)
CHEMISTRY (Theory)

*निर्धारित समय : **3** घण्टे*
*Time allowed : **3** hours*

*अधिकतम अंक : **70***
*Maximum Marks : **70***

General Instructions :

Read the following instructions carefully and follow them :

(i) *This question paper contains **33** questions. **All** questions are **compulsory**.*

(ii) *This question paper is divided into **five** sections – **Section A, B, C, D** and **E**.*

(iii) ***Section A** – questions number **1** to **16** are multiple choice type questions. Each question carries **1** mark.*

(iv) ***Section B** – questions number **17** to **21** are very short answer type questions. Each question carries **2** marks.*

(v) ***Section C** – questions number **22** to **28** are short answer type questions. Each question carries **3** marks.*

(vi) ***Section D** – questions number **29** and **30** are case-based questions. Each question carries **4** marks.*

(vii) ***Section E** – questions number **31** to **33** are long answer type questions. Each question carries **5** marks.*

(viii) *There is no overall choice given in the question paper. However, an internal choice has been provided in few questions in all the sections except Section A.*

(ix) *Kindly note that there is a separate question paper for Visually Impaired candidates.*

(x) *Use of calculator is **not** allowed.*

SECTION A

*Questions no. **1** to **16** are Multiple Choice type Questions, carrying **1** mark each.* *16×1=16*

1. With increase in atomic number, the melting point of transition metals :

(A) first increases and then decreases

(B) increases continuously

(C) first decreases and then increases

(D) remains constant

2. Which of the following aqueous solutions will have the highest osmotic pressure ?

(A) 1% KCl (B) 1% glucose

(C) 1% urea (D) 1% $CaCl_2$

3. A conductivity cell usually consists of two :

(A) Copper electrodes (B) Platinum electrodes

(C) Zinc electrodes (D) Iron electrodes

4. Which of the following is the slope of the first order reaction in the plot of ln[R] vs. time ?

(A) $+ k$ (B) $\frac{+ k}{2 \cdot 303}$

(C) $- k$ (D) $\frac{- k}{2 \cdot 303}$

5. Which of the following represents the fraction of molecules with energies equal to or greater than E_a ?

(A) $\frac{-E_a}{RT}$

(B) $e^{-E_a/RT}$

(C) $e^{+E_a/RT}$

(D) $\frac{+E_a}{RT}$

6. The number of moles of AgCl precipitated when excess $AgNO_3$ solution is mixed with one mole of $[Co(NH_3)_6]Cl_3$ is :

(A) 0 (B) 3

(C) 6 (D) 4

7. Which of the following haloalkanes is most reactive towards S_N2 reaction ?

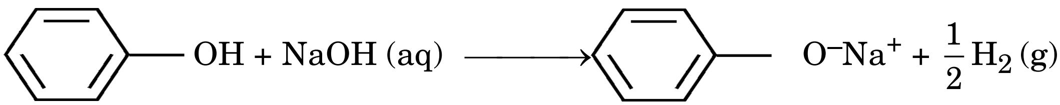
(A) $CH_3 - CH_2 - I$

(B) $CH_3 - CH_2 - Br$

(C) $CH_3 - CH_2 - Cl$

(D) $CH_3 - CH_2 - F$

8. The reaction



suggests that phenol is :

(A) Basic (B) Neutral

(C) Acidic (D) Amphoteric

9. Phenol reacts with conc. HNO_3 to form :

(A) Salicylic acid (B) Picric acid

(C) Benzoic acid (D) Phthalic acid

10. Anilinium hydrogen sulphate on heating at 453 – 473 K produces which of the following as a major product ?

(A) 2-aminobenzene sulphonic acid

(B) benzene sulphonic acid

(C) 2-aminobenzoic acid

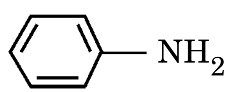
(D) sulphanilic acid

11. Which of the following amines does ***not*** give foul smell of isocyanide on heating with chloroform and ethanolic KOH ?

(A) $CH_3 - CH_2 - NH_2$

(B) $CH_3 - \underset{\substack{| \\ CH_3}}{CH} - CH_2 - NH_2$

(C) $(CH_3 - CH_2)_3N$

(D) 

12. Deficiency of vitamin B_1 causes the disease :

(A) Convulsions

(B) Beri-Beri

(C) Fissuring at corners of mouth and lips (Cheilosis)

(D) Muscular weakness

For Questions number 13 to 16, two statements are given — one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer to these questions from the codes (A), (B), (C) and (D) as given below.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is ***not*** the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

13. *Assertion (A) :* A mixture of *o*-nitrophenol and *p*-nitrophenol can be separated by fractional distillation.

Reason (R) : *o*-nitrophenol is steam volatile due to intramolecular hydrogen bonding.

14. *Assertion (A) :* Aquatic species are more comfortable in cold water than in warm water.

Reason (R) : Solubility of oxygen gas in water decreases with increase in temperature.

15. *Assertion (A) :* First ionisation enthalpy of Cr is lower than that of Zn.

Reason (R) : Zinc is a non-transition element.

16. *Assertion (A) :* Vitamin C cannot be stored in our body.

Reason (R) : Vitamin C is water soluble and excreted in urine.

SECTION B

17. (a) A and B liquids on mixing show rise in temperature. Which type of deviation from Raoult's law is there and why ?

(b) Why can azeotropic mixture not be separated by fractional distillation ? *2*

18. Why is molecularity applicable only for elementary reactions and order is applicable for elementary as well as complex reactions ? *2*

19. Write two differences between DNA and RNA. *2*

20. (a) Write the IUPAC name of the complex $[Pt(en)_2Cl_2]^{2+}$. Draw the structure of geometrical isomer of this complex which is optically inactive. *2*

OR

(b) (i) Write the formula of the following coordination compound : Pentaamminecarbonatocobalt(III)chloride

(ii) Write the IUPAC name of the linkage isomer of the complex $[Co(NH_3)_5(NO_2)]Cl_2$. *1+1=2*

21. Why are haloarenes less reactive towards nucleophilic substitution reaction ? How does the presence of nitro ($-NO_2$) group at ortho- and para-positions in haloarenes increase the reactivity towards nucleophilic substitution reaction ? *2*

SECTION C

22. (a) Write the name of the cell which is generally used in inverters. Write the reactions taking place at anode and cathode of this cell, when it is in use. *3*

OR

(b) Explain why electrolysis of an aqueous solution of NaCl gives H_2 gas at cathode and Cl_2 gas at anode ? Write overall reaction. *3*

(Given : $E^\circ_{Na^+/Na} = -2{\cdot}71$ V, $E^\circ_{H_2O/H_2} = -0{\cdot}83$ V,

$E^\circ_{Cl_2/2Cl^-} = +1{\cdot}36$ V, $E^\circ_{H^+/O_2/H_2O} = +1{\cdot}23$ V)

23. 0·3 g of acetic acid (Molar mass = 60 g mol^{-1}) dissolved in 30 g of benzene shows a depression in freezing point equal to 0·45°C. Calculate the percentage association of acid if it forms a dimer in the solution. *3*

(Given : K_f for benzene = 5·12 K kg mol^{-1})

24. A compound (A) with molecular formula C_4H_9I which is a primary alkyl halide, reacts with alcoholic KOH to give compound (B). Compound (B) reacts with HI to give (C) which is an isomer of (A). When (A) reacts with Na metal in the presence of dry ether, it gives a compound (D), C_8H_{18}, which is different from the compound formed when n-butyl iodide reacts with sodium. Write the structures of (A), (B), (C) and (D). Write the chemical equation when compound (A) is reacted with alcoholic KOH. *3*

25. The following data were obtained during the first order thermal decomposition of N_2O_5 (g) at constant volume : *3*

$$2N_2O_5\,(g) \longrightarrow 2N_2O_4\,(g) + O_2\,(g)$$

S.No.	Time/s	Total Pressure/atm
1	0	0·5
2	100	0·625

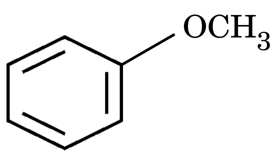
Calculate rate constant.

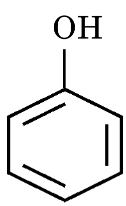
[Given : log 2 = 0·3010, log 10 = 1]

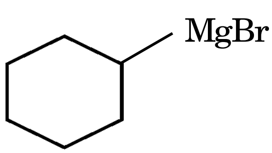
26. Write the reaction of D-Glucose with the following : *3×1=3*

(a) HCN (b) Br_2 water (c) $(CH_3CO)_2O$

27. Write structure of the products of the following reactions : *3×1=3*

(a)  + HI $\longrightarrow$

(b)  + conc. HNO_3 $\longrightarrow$

(c)  + HCHO $\xrightarrow{H^+/H_2O}$

28. Give reasons for the following : *3×1=3*

(a) Benzoic acid does not undergo Friedel-Crafts reaction.

(b) HCHO is more reactive than CH_3CHO towards addition of HCN.

(c) Vinyl group directly attached with carboxylic acid should decrease the acidity of corresponding carboxylic acid due to resonance, but on the contrary it increases the acidity.

SECTION D

The following questions are case-based questions. Read the case carefully and answer the questions that follow.

29. The Crystal Field Theory (CFT) of coordination compounds is based on the effect of different crystal fields (provided by the ligands taken as point charges) on the degeneracy of d-orbital energies of the central metal atom/ion. The splitting of the d-orbitals provides different electronic arrangements in strong and weak crystal fields. In tetrahedral coordination entity formation, the d-orbital splitting is smaller as compared to the octahedral entity.

Answer the following questions :

(a) On the basis of CFT, explain why $[Ti(H_2O)_6]Cl_3$ complex is coloured ? What happens on heating the complex $[Ti(H_2O)_6]Cl_3$? Give reason. *2*

[Atomic no. : Ti = 22]

(b) (i) What is crystal field splitting energy ? *1*

OR

(b) (ii) On the basis of Δ_o and P (pairing energy), how can you differentiate between a strong field ligand and a weak field ligand ? *1*

(c) Why are low spin tetrahedral complexes rarely observed ? *1*

30. Amines are usually formed from amides, imides, halides, nitro compounds, etc. They exhibit hydrogen bonding which influences their physical properties. In alkyl amines, a combination of electron releasing, steric and H-bonding factors influence the stability of the substituted ammonium cations in protic polar solvents and thus affect the basic nature of amines. Alkyl amines are found to be stronger bases than ammonia. Amines being basic in nature, react with acids to form salts. Aryldiazonium salts, undergo replacement of the diazonium group with a variety of nucleophiles to produce aryl halides, cyanides, phenols and arenes.

Answer the following questions :

(a) How can you convert the following ? *2*

(i) Ethanoic acid to methanamine

(ii) Propanenitrile to 1-aminopropane

(b) Why is pK_b value of aniline more than that of methylamine ? *1*

(c) (i) Arrange the following in increasing order of their basic strength in aqueous solution : *1*

$$CH_3 - NH_2, \quad (CH_3)_2NH, \quad (CH_3)_3N$$

OR

(c) (ii) Give the structures of A and B in the following reaction : *1*

$$C_6H_5NO_2 \xrightarrow{Fe/HCl} A \xrightarrow[273\,K]{HNO_2} B$$

SECTION E

31. (a) (i) An organic compound (X) having molecular formula $C_5H_{10}O$ can show various properties depending on its structures. Draw each of the structures if it :

(I) shows Cannizzaro reaction. *1*

(II) reduces Tollens' reagent and has a chiral carbon. *1*

(III) gives positive iodoform test. *1*

(ii) Write the reaction involved in the following : 2

(I) Clemmensen reduction

(II) Etard reaction

OR

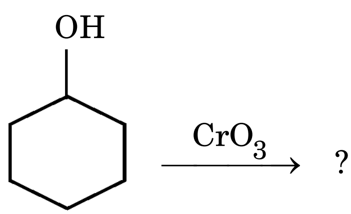
(b) Answer the following questions : $5 \times 1 = 5$

(i) Draw structure of the methyl hemiacetal of methanal.

(ii) There are two $-NH_2$ groups in semicarbazide. However only one is involved in the formation of semicarbazones. Give reason.

(iii) How will you convert ethanol to 3-hydroxybutanal ?

(iv) Complete the following equation :



(v) Write the final product formed when phthalic acid is treated with NH_3 followed by strong heating.

32. (a) (i) Calculate the emf of the following cell at 25°C : 3

$Zn(s) \mid Zn^{2+}\ (0.1\ M) \parallel H^+\ (0.01\ M) \mid H_2(g)\ (1\ bar), Pt(s)$

[Given : $E^\circ_{Zn^{2+}/Zn} = -0.76$ V, $E^\circ_{2H^+/H_2} = 0.00$ V, log 10 = 1]

(ii) State Faraday's second law of electrolysis. How much electricity is required in terms of Faraday for the reduction of 1 mol of $Cr_2O_7^{2-}$ to Cr^{3+} ? 2

OR

(b) Answer the following questions :

(i) The conductivity of 0·20 M solution of KCl is $2{\cdot}48 \times 10^{-2}$ S cm^{-1}. Calculate its molar conductivity and degree of dissociation (α).

[Given : $\lambda^{\circ}_{(K^+)} = 73{\cdot}5$ S cm^2 mol^{-1}

$\lambda^{\circ}_{(Cl^-)} = 76{\cdot}5$ S cm^2 mol^{-1}]

(ii) Calculate $\Delta_r G^{\circ}$ of the following cell :

$Mg(s) + Cu^{2+}(aq) \longrightarrow Mg^{2+}(aq) + Cu(s)$

[Given : $E^{\circ}_{Mg^{2+}/Mg} = -\ 2{\cdot}37$ V, $E^{\circ}_{Cu^{2+}/Cu} = +\ 0{\cdot}34$ V

1 F = 96500 C mol^{-1}]

(iii) What type of cell is mercury cell ? Why is it more advantageous than dry cell ? *2+2+1=5*

33. (a) (i) Account for the following :

(I) The $E^{\circ}_{Mn^{2+}/Mn}$ value for manganese is highly negative, whereas $E^{\circ}_{Mn^{3+}/Mn^{2+}}$ is highly positive. *1*

(II) Actinoids show wide range of oxidation states. *1*

(III) Transition metals have high melting points. *1*

(ii) Complete the following ionic equations :

(I) $5SO_3^{\ 2-} + 2MnO_4^{\ -} + 6H^+ \longrightarrow$ *1*

(II) $2MnO_4^{\ -} + H_2O + I^- \longrightarrow$ *1*

OR

(b) Answer the following questions : *5×1=5*

(i) Name two elements of 3d series for which the third ionisation enthalpies are quite high.

(ii) Out of $KMnO_4$ and K_2MnO_4, which one is paramagnetic and why ?

(iii) Write any one consequence of lanthanoid contraction.

(iv) How do you prepare potassium manganate from pyrolusite ore ?

(v) Why is the ability of oxygen more than fluorine to stabilise higher oxidation states of transition metals ?